Feature Engineering

Feature Scaling

Normalization

Janek
175 lbs. 5,9 ft
T-shirt: L

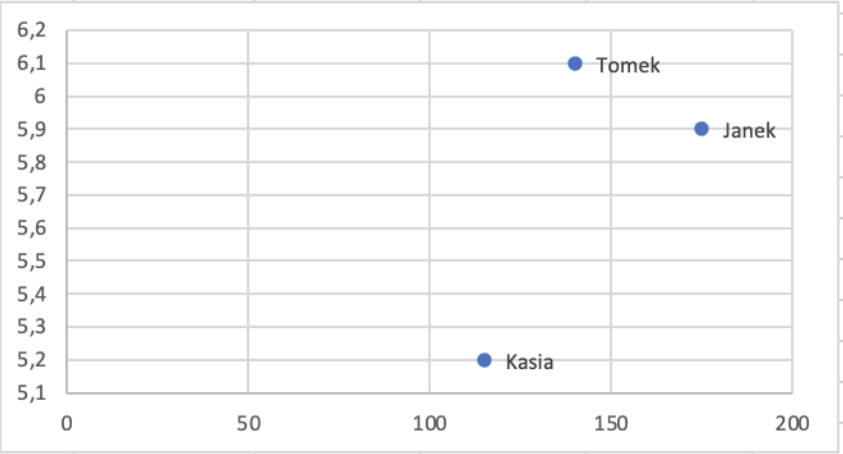
Kasia
115 lbs, 5.2 ft
T-shirt: S

Tomek
140 lbs, 6.1 ft
T-shirt: ???

Feature Scaling

Normalization

	weight (lbs)	height (ft)
Janek	175	5,9
Kasia	115	5,2
Tomek	**140**	**6,1**
Distance		
Tomek <> Kasia	25,02	
Tomek <> Janek	35,00	



Tomek is closer to Kasia:
Algorytm will recommend
a **T-shirt S**

Feature Scaling

Normalization

$$X_n = \frac{X - X_{min}}{X_{max} - X_{min}}$$

	weight (lbs)	height (ft)	weight_n	ht_n
Janek	175	5,9	1	0,78
Kasia	115	5,2	0	0,00
Tomek	**140**	**6,1**	0,42	1,00
Distance				
Tomek <> Kasia	1,08			
Tomek <> Janek	0,62			

Tomek is closer to Janek:
Algorytm will recommend
a **T-shirt L**

Feature Scaling

Normalization

In [38]:
```
import pandas as pd
import seaborn as sns
from sklearn.preprocessing import MinMaxScaler, StandardScaler
```

In [3]:
```
df = pd.read_csv('weight_height.csv')
```

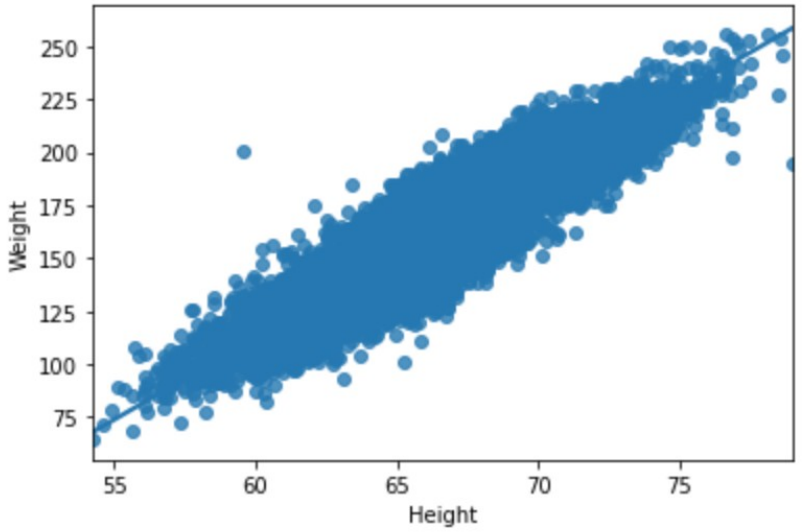
In [7]:
```
df.head()
```

Out[7]:

	Gender	Height	Weight
0	Male	73.847017	241.893563
1	Male	68.781904	162.310473
2	Male	74.110105	212.740856
3	Male	71.730978	220.042470
4	Male	69.881796	206.349801

In [36]:
```
sns.regplot(x=df["Height"], y=df["Weight"])
```

Out[36]:
```
<AxesSubplot:xlabel='Height', ylabel='Weight'>
```



Feature Scaling

Normalization

$$X_n = \frac{X - X_{min}}{X_{max} - X_{min}}$$

In [31]:
```
scaler = MinMaxScaler()
```

In [32]:
```
scaler.fit(df[['Height']])
df['H_n'] = scaler.transform(df[['Height']])
```

In [33]:
```
scaler.fit(df[['Weight']])
df['W_n'] = scaler.transform(df[['Weight']])
```

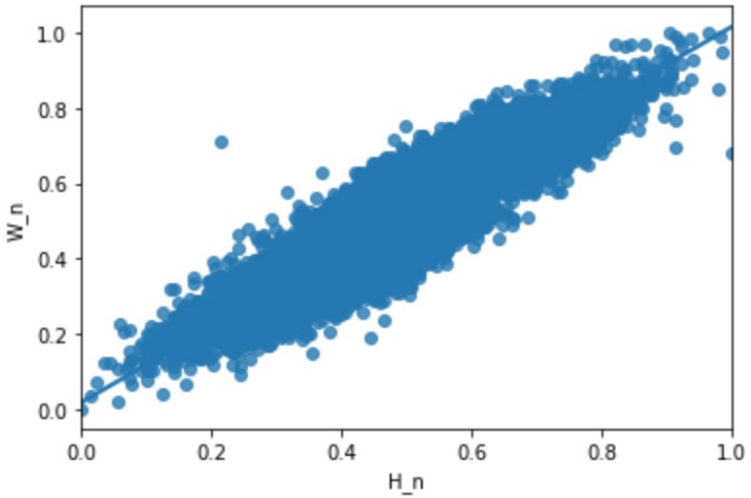
In [34]:
```
df.head()
```

Out[34]:

	Gender	Height	Weight	H_n	W_n
0	Male	73.847017	241.893563	0.791728	0.926922
1	Male	68.781904	162.310473	0.586958	0.510613
2	Male	74.110105	212.740856	0.802364	0.774421
3	Male	71.730978	220.042470	0.706182	0.812616
4	Male	69.881796	206.349801	0.631424	0.740988

In [35]:
```
sns.regplot(x=df["H_n"], y=df["W_n"])
```

Out[35]: <AxesSubplot:xlabel='H_n', ylabel='W_n'>



Feature Scaling

Standardization

$$X_s = \frac{X - \mu}{\sigma}$$

μ = mean

σ = standard deviation

```
In [39]: scaler = StandardScaler()
```

```
In [43]: scaler.fit(df[['Height']])
         df['H_s'] = scaler.transform(df[['Height']])
```

```
In [44]: scaler.fit(df[['Weight']])
         df['W_s'] = scaler.transform(df[['Weight']])
```

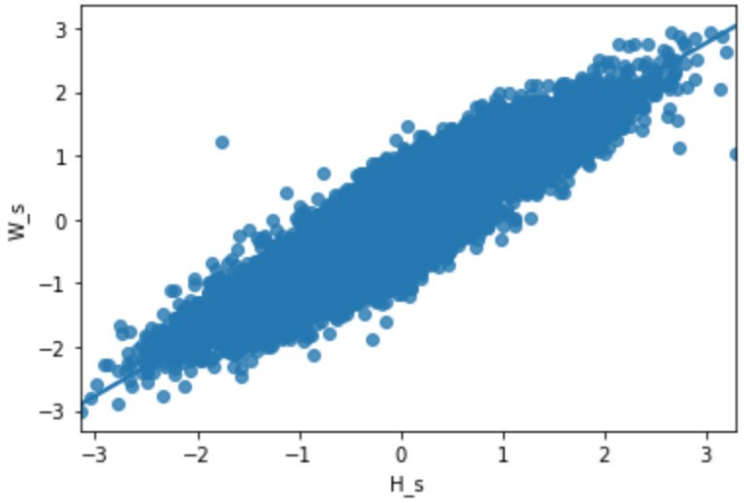
```
In [45]: df.head()
```

Out[45]:

	Gender	Height	Weight	H_n	W_n	H_s	W_s
0	Male	73.847017	241.893563	1.944061	2.507852	1.944061	2.507852
1	Male	68.781904	162.310473	0.627537	0.027223	0.627537	0.027223
2	Male	74.110105	212.740856	2.012443	1.599153	2.012443	1.599153
3	Male	71.730978	220.042470	1.394060	1.826747	1.394060	1.826747
4	Male	69.881796	206.349801	0.913421	1.399942	0.913421	1.399942

```
In [46]: sns.regplot(x=df["H_s"], y=df["W_s"])
```

Out[46]: <AxesSubplot:xlabel='H_s', ylabel='W_s'>

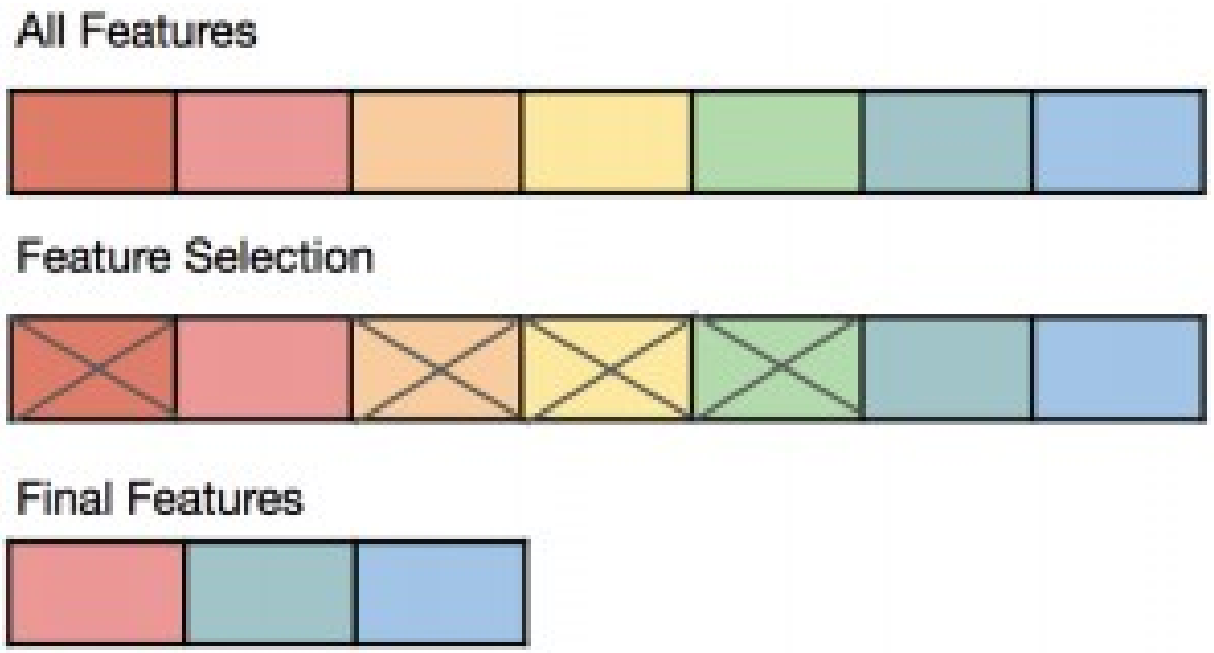


Feature Selection

Feature selection is the process of

1. identifying and selecting a subset of input variables that are most relevant to the target variable

2. reducing the number of input variables when developing a predictive model.



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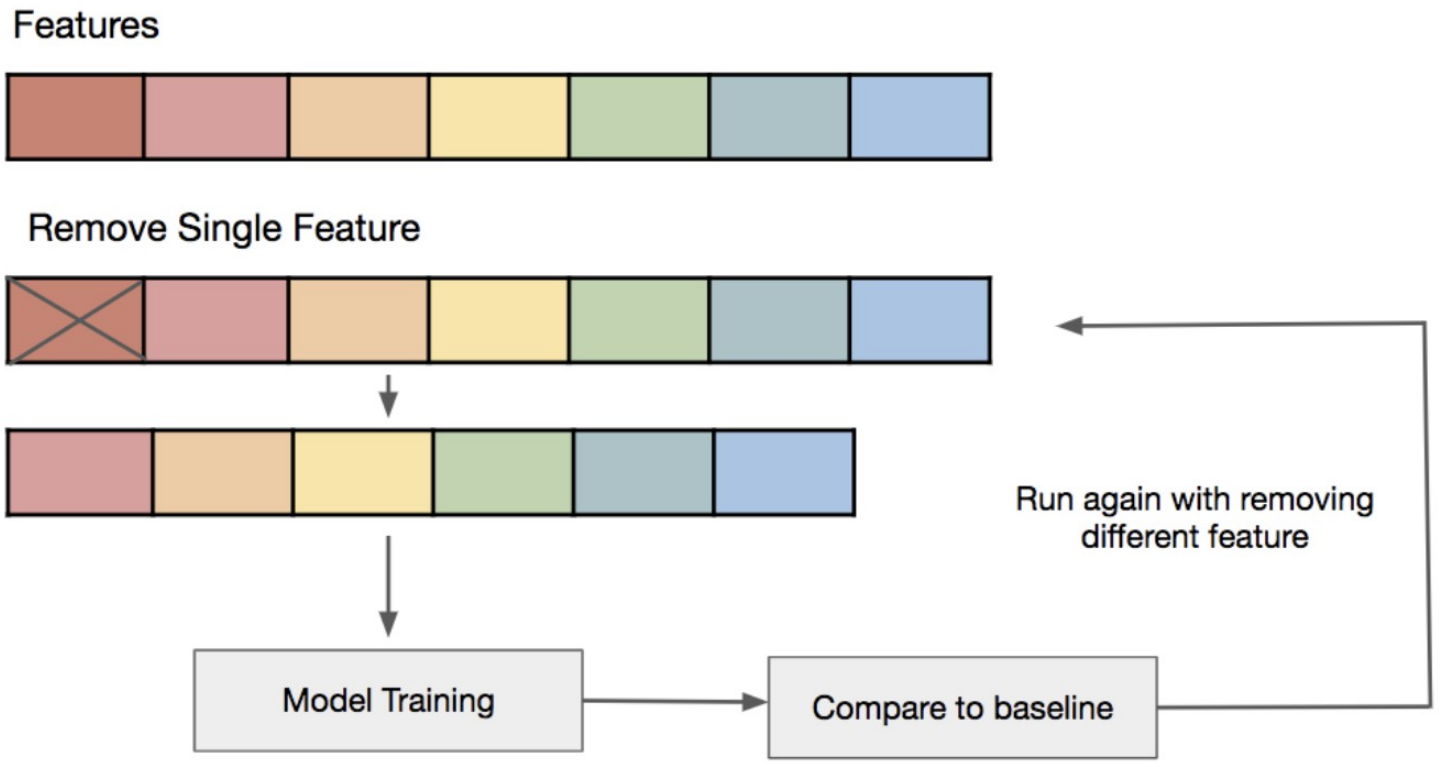

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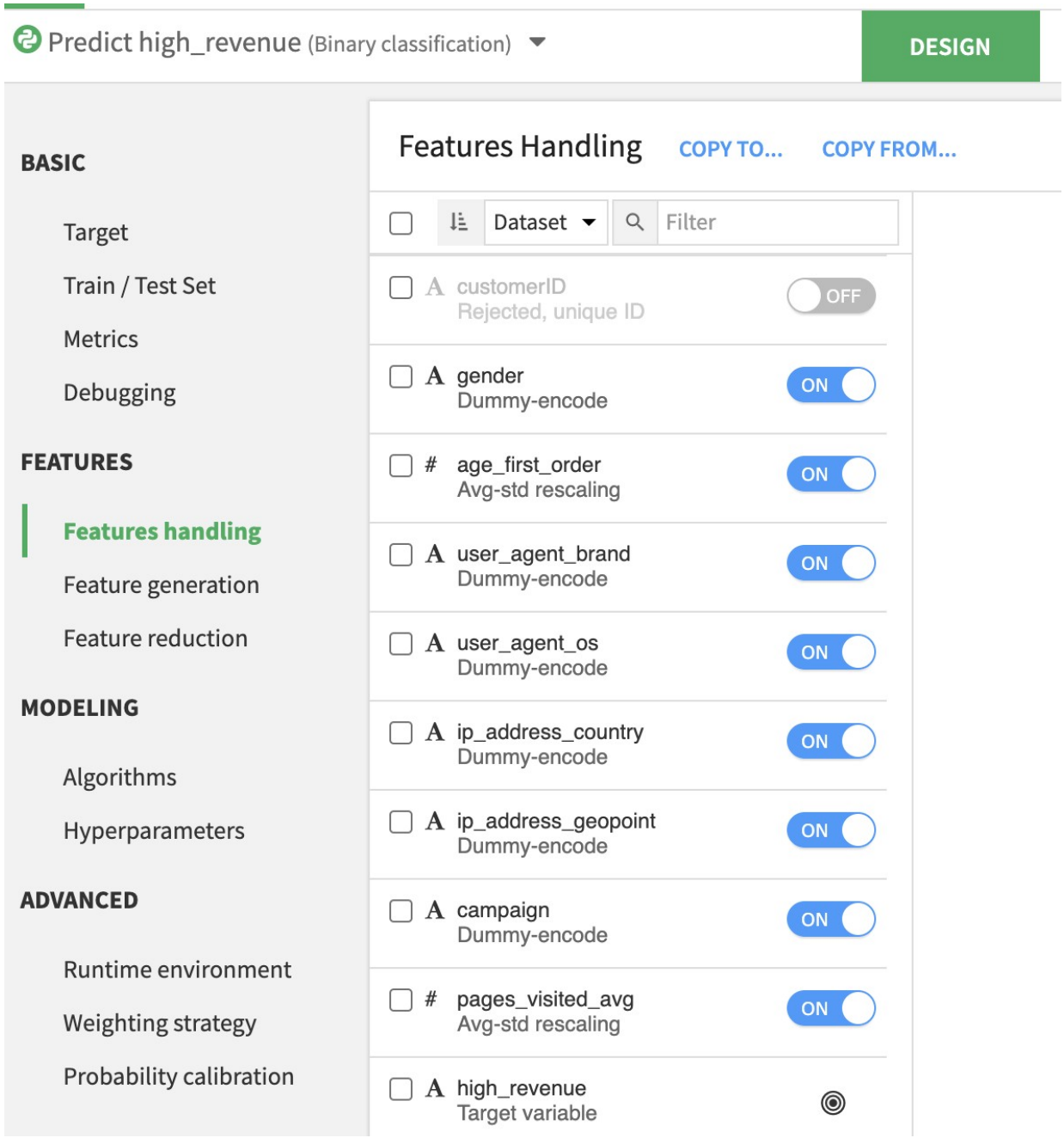

Feature Selection

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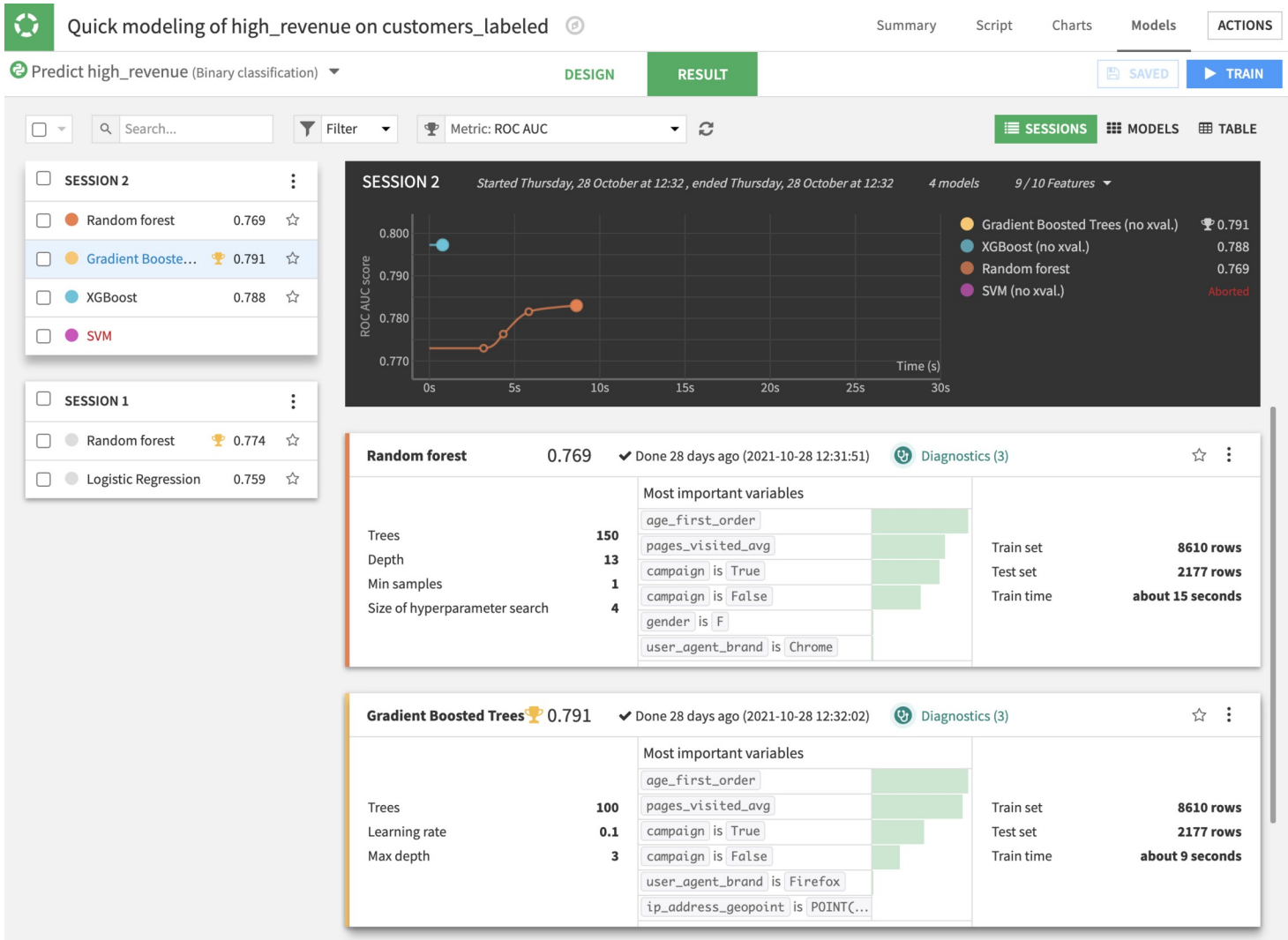
2. reducing the number of input variables when developing a predictive model.

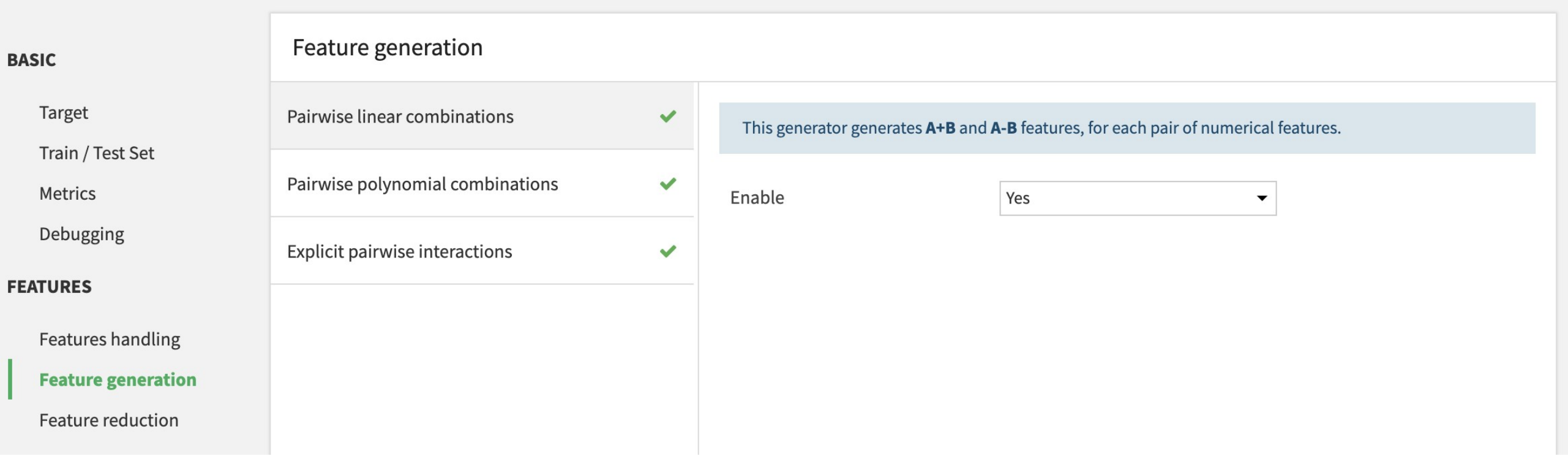


Feature generation

Pairwise:

A+B

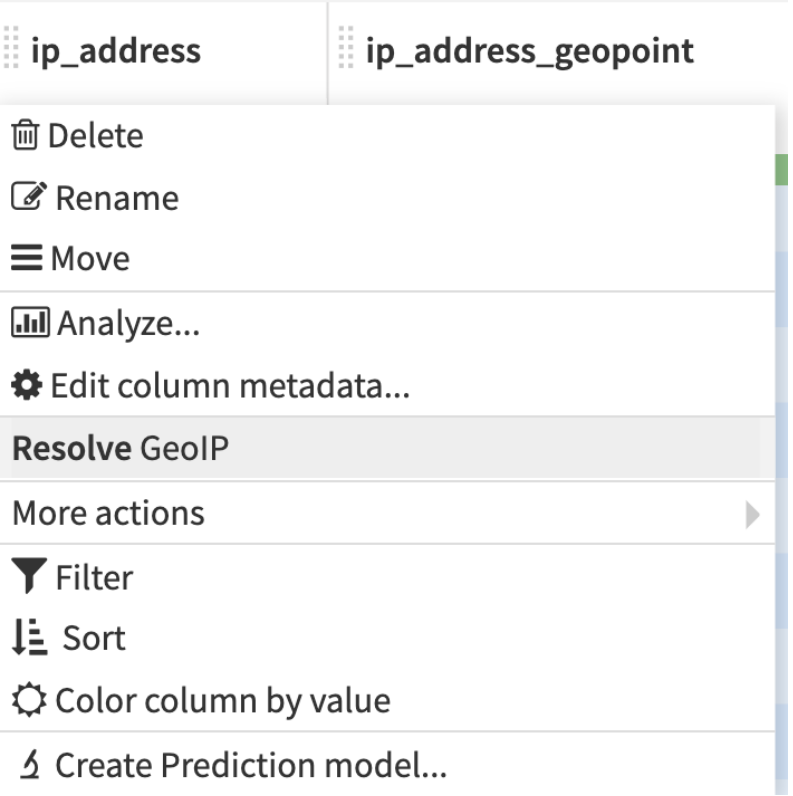
A*B



https://knowledge.dataiku.com/latest/courses/machine-learning/tune-model/concept-feature-generation-reduction.html

Feature generation

Data Enrichment



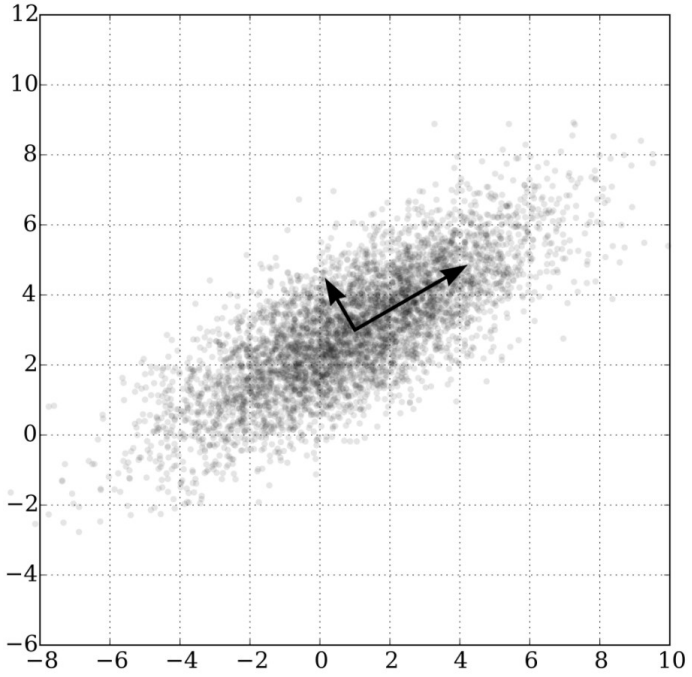
ip_address	ip_address_geopoint	ip_address_city	ip_address_region_hierarchy
IP address	GeoPoint	Text	Text
69.8.136.110	POINT(-97.822 37.751)		
67.74.254.174	POINT(-97.822 37.751)		
95.135.70.206	POINT(30.5233 50.45)		
129.50.205.26	POINT(-97.822 37.751)		
81.98.234.29	POINT(-2.4578 53.7683)	Blackburn	England,Blackburn with Darwen
76.177.15.150	POINT(-84.2955 37.7546)	Richmond	Kentucky
174.83.130.76	POINT(-119.6115 38.8703)	Gardnerville	Nevada
125.128.130.195	POINT(126.9783 37.5985)	Seoul	Seoul
137.155.219.158	POINT(-76.4967 37.0768)	Newport News	Virginia
126.217.132.156	POINT(138.8333 35.8)	Ochiai	Yamanashi
42.226.241.79	POINT(113.5325 34.6836)		Henan
108.245.154.218	POINT(-84.7747 37.6465)	Danville	Kentucky

Feature reduction

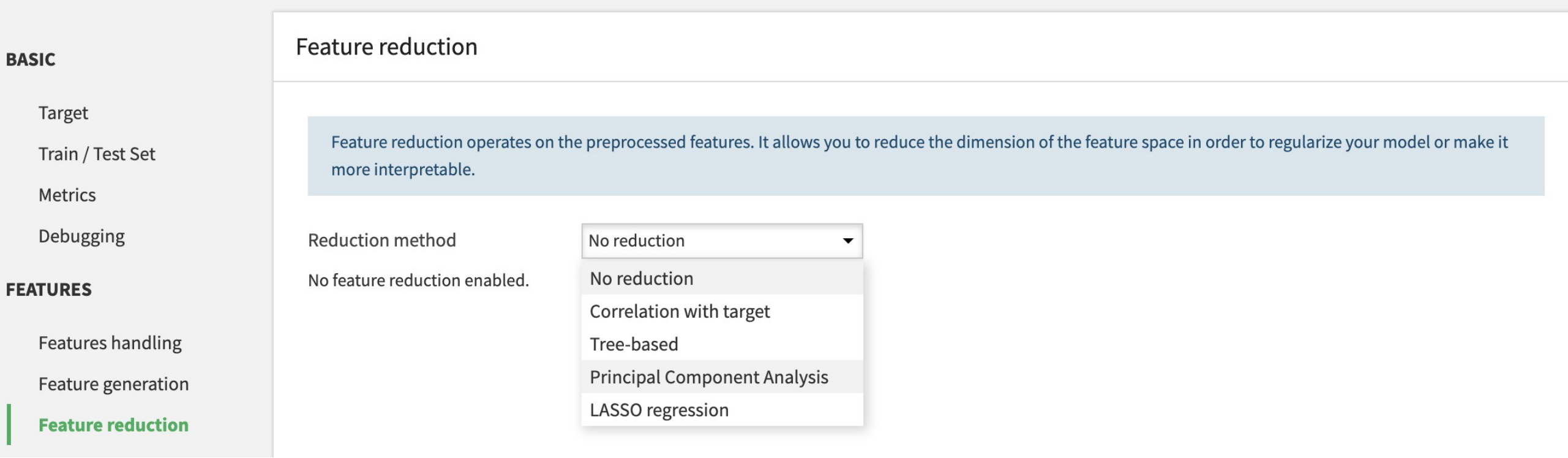
PCA – Principal Component Analysis

Why?

- Easier to plot
- Easier to understand
- Faster computations
- … and different conclusions.



Feature reduction



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Feature Engineering and Selection (Book Review)

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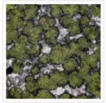

Discover Feature Engineering, How to Engineer Features and How to Get Good at It

by **Jason Brownlee** on September 26, 2014 in **Data Preparation**

Feature engineering is an informal topic, but one that is absolutely known and agreed to be key to success in [...]

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Discover Feature Engineering, How to Engineer Features and How to Get Good at It

by **Jason Brownlee** on September 26, 2014 in **Data Preparation**

Tweet Share Share

Last Updated on August 15, 2020

Feature engineering is an informal topic, but one that is absolutely known and agreed to be key to success in applied machine learning.

In creating this guide I went wide and deep and synthesized all of the material I could.

You will discover what feature engineering is, what problem it solves, why it matters, how to engineer features, who is doing it well and where you can go to learn more and get good at it.

If you read one article on feature engineering, I want it to be this one.

> *feature engineering is another topic which doesn't seem to merit any review papers or books, or even chapters in books, but it is absolutely vital to ML success. […] Much of the success of machine learning is actually success in engineering features that a learner can understand.*

— Scott Locklin, in "Neglected machine learning ideas"

Kick-start your project with my new book Data Preparation for Machine Learning, including *step-by-step tutorials* and the *Python source code* files for all examples.

Let's get started.

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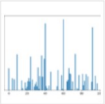

How to Perform Feature Selection for Regression Data

by **Jason Brownlee** on June 8, 2020 in **Data Preparation**

Feature selection is the process of identifying and selecting a subset of input variables that are most relevant to the […]

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How to Perform Feature Selection With Numerical Input Data

by **Jason Brownlee** on June 5, 2020 in **Data Preparation**

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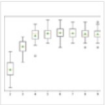

Recursive Feature Elimination (RFE) for Feature Selection in Python

by **Jason Brownlee** on May 25, 2020 in **Data Preparation**

Recursive Feature Elimination, or RFE for short, is a popular feature selection algorithm. RFE is popular because it is easy […]

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by **Jason Brownlee** on November 27, 2019 in **Data Preparation**

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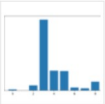

How to Perform Feature Selection with Categorical Data

by **Jason Brownlee** on November 25, 2019 in **Data Preparation**

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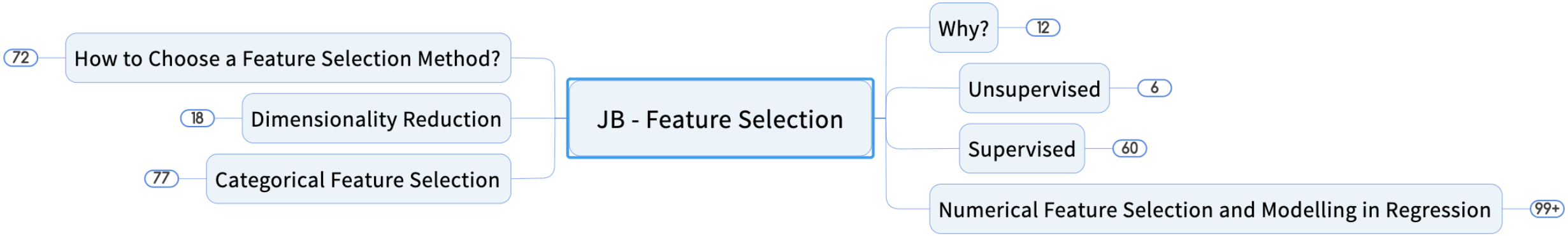
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Dimensionality Reduction
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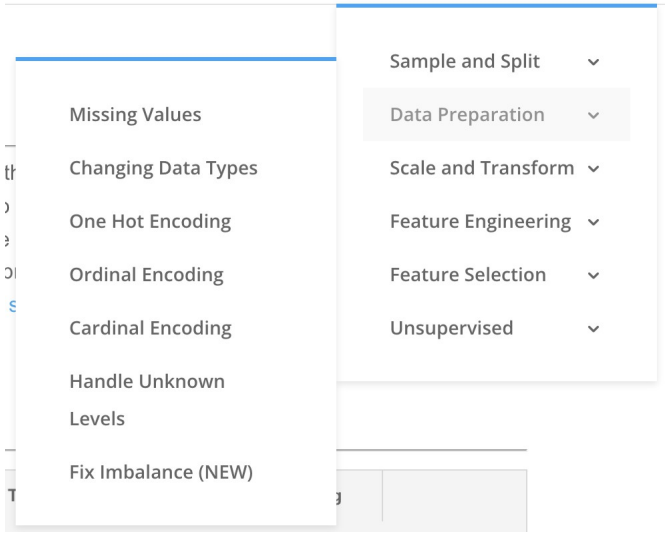
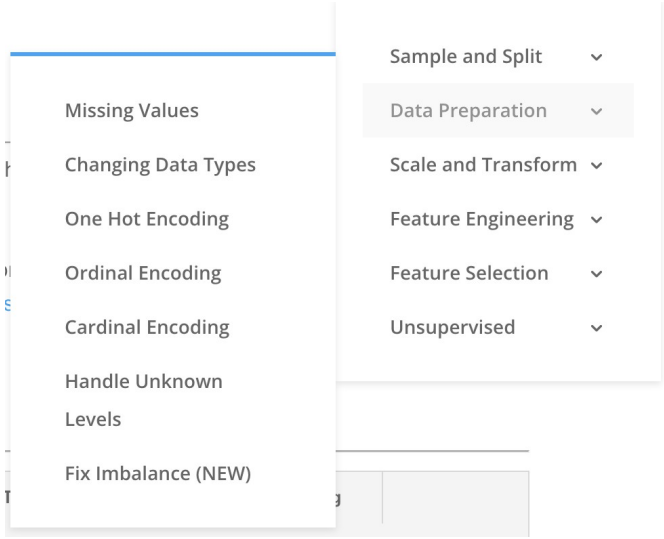
Numerical Feature Selection and Modelling in Regression
99+

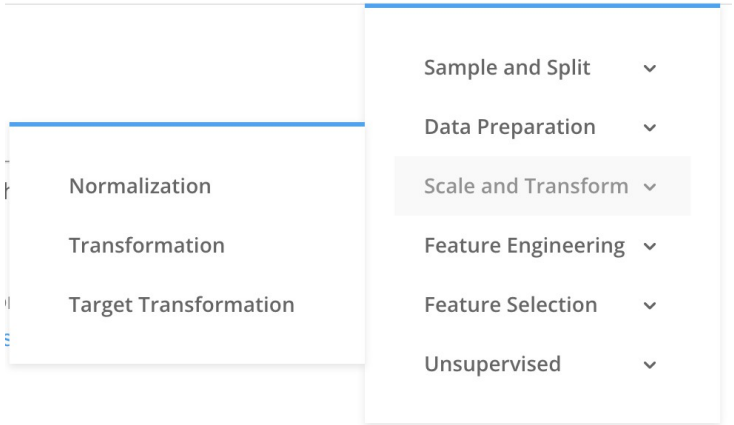
Tools

Pycaret

https://pycaret.org/preprocessing/

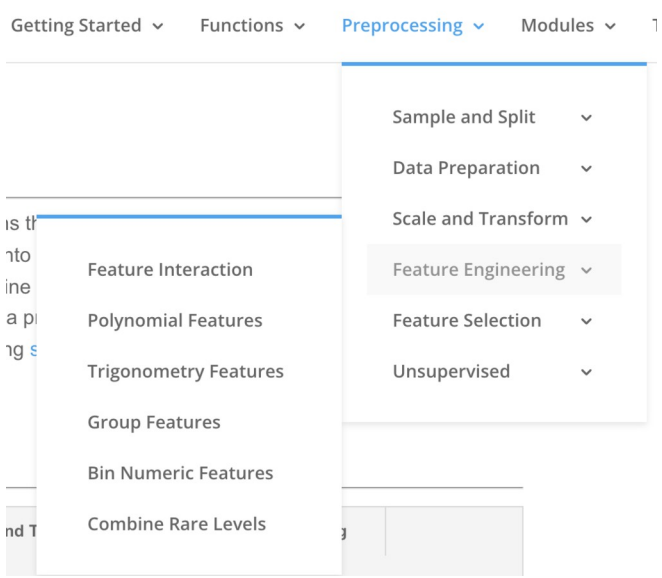


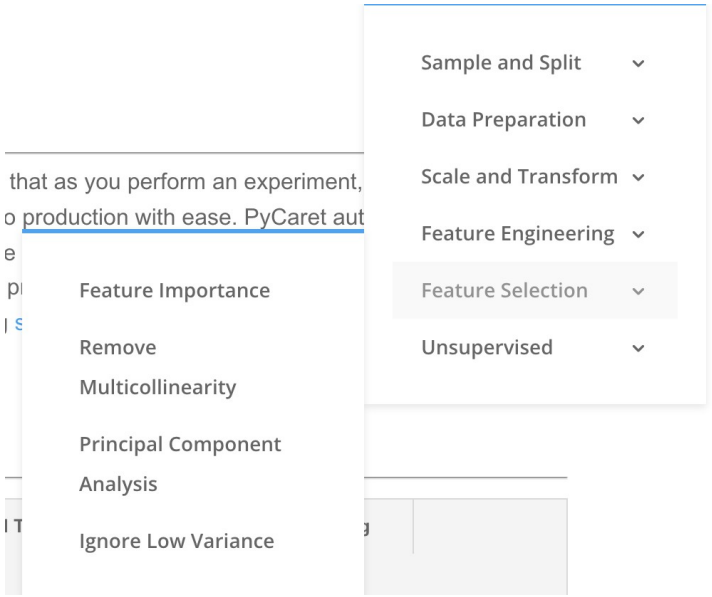


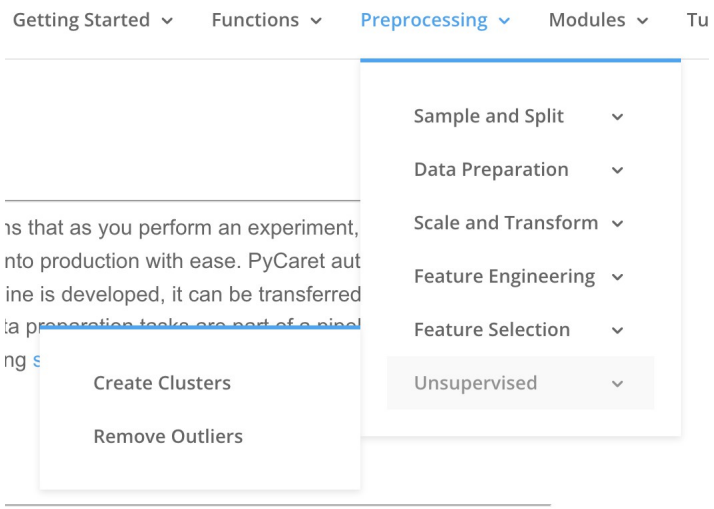


Pycaret

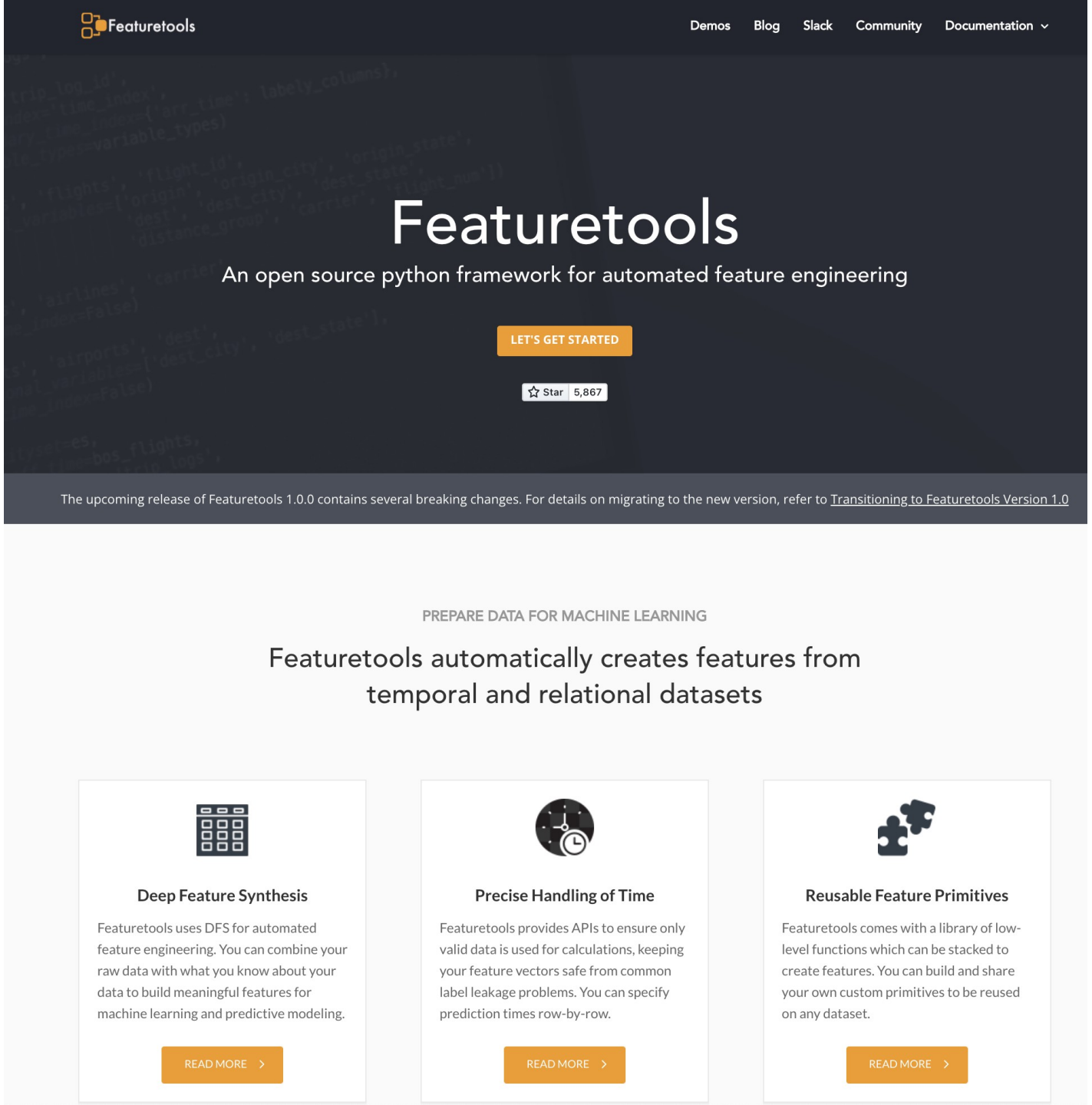
https://pycaret.org/preprocessing/







Featuretools.com



Featuretools.com

Demos

The end-to-end demos below exhibit how you might use Featuretools in real-world applications. If there's a use case you'd like to see, let us know.

All | Ecommerce | Education | Finance | Health Care | Insurance | IOT | Retail | Sports | Transportation

Predict Next Purchase

Predict Remaining Useful Life

Predict Appointment No-Show

Predict Loan Repayment

Predict Correct Answer

Predict Olympic Medals

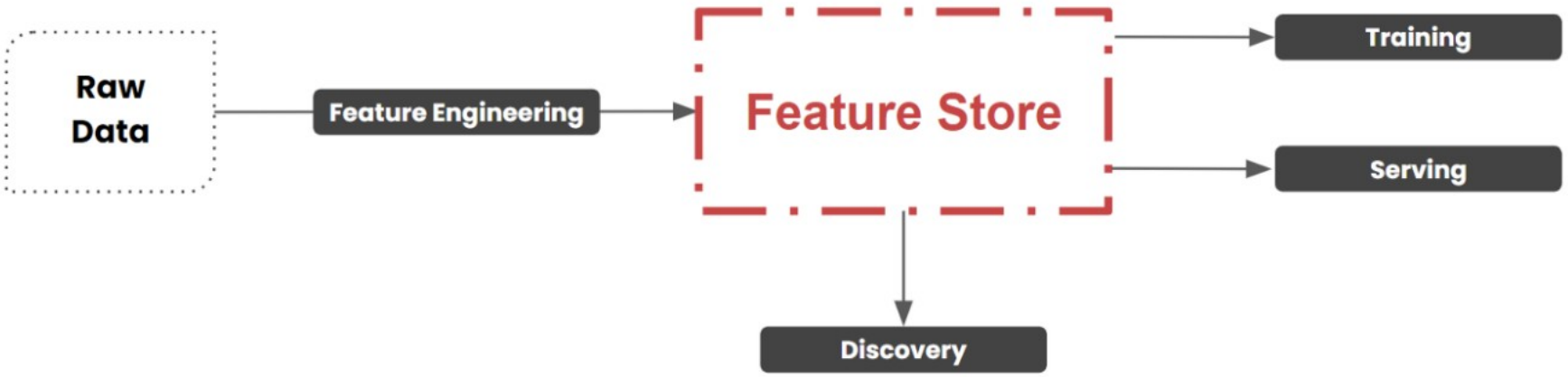
Predict Customer Churn

Predict Taxi Trip Duration

Predict Household Poverty

Predict Malicious Internet Traffic

Feature stores



https://towardsdatascience.com/feature-store-as-a-foundation-for-machine-learning-d010fc6eb2f3

Feature stores

Feast 0.15 has just been released! Check it out →

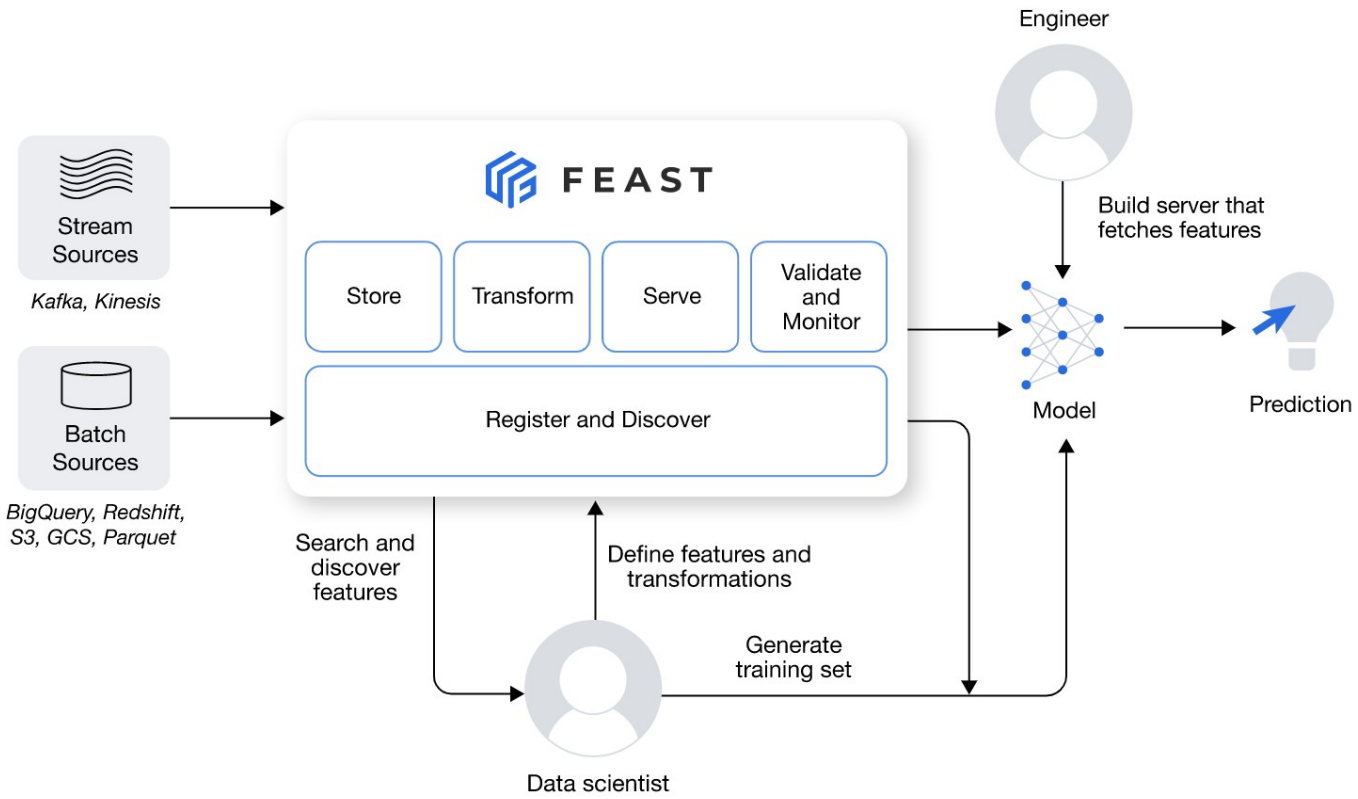
FEAST

Star 2,496 Resources Blog Docs

Serve your features in production

Feast is an open-source feature store. It is the fastest path to operationalizing analytic data for model training and online inference.

Get started



Feature stores

HOPSWORKS

Feature Store | Open Source | Pricing | Community | Resources

START NOW

Full AI lifecycle for MLOps, built around the industry leading Feature Store.

demo_fs_meb10000

Search for Ctrl + P

Admin Admin

Home

Features Store

Feature Groups

Training Datasets

Storage Connectors

Compute

Jobs

Configuration

Project settings

Back to current Hopsworks

Set up storage connector

Name

name of the source

Description optional

description

Protocol

AWS S3 | JDBC | HopsFS | Redshift | Azure Data Lake | Snowflake

Parameters

Connection URL

account_name.snowflakecomputing.com

User

user

Authentication

Password | Token

Password